



- An inspector rejects 0.08% of the sample, those being directive. How many samples should be inspect to reject 2 of them: **(Percentage) NIMCET-2007**
(a)2000 (b)2500 (c)2250 (d)none
- In an examination there are 100 questions divided into 3 parts A, B and C and each part should contain atleast one question. Each question in parts A, B and C carry 1, 2 and 3 marks respectively. Part A is for atleast 60% of the total marks and part B should contain 23 questions. How many questions must be set in part C? **(Percentage) NIMCET-2015**
(a)1 (b)2
(c)3 (d)Cannot be determined
- In a city, 40% of the adults are illiterate while 85% of the children are literate. If the ratio of the adults to that of the children is 2: 3, then what per cent of the population is literate? **(Percentage) NIMCET-2018**
(a)20% (b)25% (c) 50% (d) 75%
- Fresh grapes contain 90% by weight while dried grapes contain 20% water by weight. What is the weight of dry grapes available from 20 kg of fresh grapes?
(a)2.5 kg (b)2.4 kg (c)2 kg (d)10 kg
(Percentage) NIMCET-2019
- In a city, 40.1% of the adults are illiterate while 85.1% the children are literate. If the ratio of the adults to that the children is 2:3, then what percent of the population is literate?
(Percentage) NIMCET-2020
(a)20% (b)25% (c)50% (d)75%
- In a competitive examination in Maharashtra state 9% candidates got selected from the total appeared candidates. Tripura state had an equal number candidates appeared and 12% candidates got selected with 102 more candidates got selected than Maharashtra state. What was the number of candidates appeared from each state? **(Percentage) NIMCET-2020**
(a)3400 (b)3000 (c)2850 (d)3200
- Shiva gave 40% of his monthly salary to his mother from the remaining he used 7% for electronic gadgets and 23% he kept aside for his monthly expenses. The remaining amount he transferred to his friend's account. The sum of the amount he gave to his mother and he

transferred to his friend account was ₹ 41000. What was Shiva's monthly salary? **(Percentage) NIMCET-2020**

(a) ₹ 50500 (b) ₹ 49000 (c) ₹ 50000 (d) ₹ 45000

- In an examination, 78% of the total students who appeared were successful. If the total number of failures was 176 and 34% got first class, then how many students got first class?

(Percentage) NIMCET-2020

(a)272 (b)112 (c)210 (d)254

- If the numerator of a fraction is increased by 25% and denominator decreased by 20%, the new value is $\frac{5}{4}$. What is the original value? **(percentage) NIMCET-2020**
(a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{7}{8}$ (d) $\frac{3}{7}$

- In the half – yearly exam only 60% of the students were passed.

Out of these (passed in half- yearly) only 70% students are passed in annual exam, out of remaining students (who fail in half- yearly exam) 80% passed in annual exam. What per cent of the students passed the annual exam? **(Percentage) NIMCET-2023**

(a)72% (b)76% (c)65% (d)74%

- Hemant deposits 10% of his salary in PF. He saves 30% of the remaining salary. The ratio of his expense on medicine and groceries is 3: 4 of the remaining salary after saving. If his expense on the medicine was ₹2700, then find his monthly salary. **(Percentage) NIMCET-2023**

(a) ₹30000 (b) ₹20000 (c) ₹10000 (d) ₹15000

ANSWER

1	b	2	a	3	d	4	a	5	d
6	a	7	c	8	a	9	b	10	d
11	c								

SOLUTION:

- (b)** Inspector rejects 0.08%

let there be x mater.

No. if defective maters = 0.08% of x

$$2 = \frac{0.08 \times x}{100}$$

$$x = \frac{200}{0.08}, x = 2500$$

option (b)

- (a)** Suppose, the number of questions in group A, B and C be x , y , and z , respectively.
Then, $x + y + z = 100$ (given)





Total marks would be $x + 2y + 3z$

Given, $y = 23$

Total marks from section B = 46

Different possible values for z are 1, 2, 3.....

$\therefore$ Corresponding values for x are 76, 75, 74...

(Since, total questions are 100)

when $z = 1$, $x = 76$, $y = 23$

Total marks from groups A, B and C are 76, 46, 3 respectively.

Percentage marks from A = $\frac{76}{76 + 46 + 3} = \frac{76}{125} < 60\%$

when, $z = 2$, $x = 75$, $y = 23$

Percentage marks from A = $\frac{75}{76 + 46 + 6} = \frac{75}{127} < 60\%$

For all other values, when z increases, x decreases and contribution of marks from A keeps decreasing.

$\therefore$ There is only 1 possible value for questions from group C.

3. (d) Let the total population be 100.

$\therefore$ Number of adults = $\frac{2}{5} \times 100 = 40$

and number of children = 60

Now, 60% of adults are literate.

75×5

$\therefore$ Number of literate adults = $\frac{60}{100} \times 40 = 24$

and number of literate children = $\frac{85}{100} \times 60 = 51$

$\therefore$ Total number of literates = $24 + 51 = 75$

$\therefore$ Required percentage = $\frac{75}{100} \times 100 = 75\%$

4. (a) 90% of the weight of the fresh grapes is water, so 10% is not water.

$10\% \times 20 \text{ kg} = 2 \text{ kg}$ of total waterless grapes. Dried grapes contain

20% water. So, 80% of the weight of dried grapes is not water

80% dried grapes weight = 2 kg

so, dried grape weight = $\frac{2 \text{ kg}}{80\%} = \frac{2}{80} \times 100 = 25 \text{ kg}$

According to given question diagram is as following

5. (d) Let the number of adults is $2x$ and number of children be $3x$.

Number of literate adults = 60% of $2x = \frac{6x}{5}$

Number of literate children = 85% of $3x = \frac{51x}{20}$

Total literate population = $\frac{6x}{5} + \frac{51x}{20} = \frac{75x}{20}$

% of literate = $\frac{75x}{5x} \times 100$

$\Rightarrow 75\%$

6. (a) Given,

$\Rightarrow$ Selected candidate in Maharashtra from the total appeared candidates = 9%

$\Rightarrow$ Selected candidate in Tripura from the total appeared candidates = 12%

$\Rightarrow$ In Tripura number of selected candidate more than that of Maharashtra = 102

Let total number of candidates from each state be 100 unit.

Selected candidate in Maharashtra from the total appeared candidates = $\left(\frac{9}{100}\right) \times 100 = 9 \text{ Unit}$

$\Rightarrow$ Selected candidate in Tripura from the total appeared candidates = $\left(\frac{12}{100}\right) \times 100 = 12 \text{ Unit}$

In Tripura number of selected candidates more than that of Maharashtra = 102

$\Rightarrow (12-9) \text{ unit} = 102$

$\Rightarrow 3 \text{ unit} = 102$

$\Rightarrow 1 \text{ unit} = 102/3$

$\Rightarrow 1 \text{ unit} = 34$

$(1 \times 100) \text{ unit} = 34 \times 100$

$\Rightarrow 100 \text{ unit} = 3400$

$\therefore$ The number of candidates appeared from each state was 3400.

$\therefore$ The correct answer is option (a).

7. (c) Salary given to his mother = 40% = $2/5$ from the remaining, he used for electronic gadgets and monthly expenses = $(7 + 23)\% = 30\% = 3/10$

Sum of the amount he gave to his mother and transferred to his friend accounts = Rs. 41000

Let the sum of amount be 50 unit (where 50 = product of denominator of given fraction 5×10)

Salary given to his mother = $\left(\frac{2}{5} \times 50\right)$

= 20 unit

Remaining salary after giving to his mother = $50 - 20 = 30 \text{ unit}$.

Electronics gadgets and monthly expenses from the remaining =

$\left(\frac{3}{10} \times 30\right) = 9 \text{ unit}$

Remaining amount which transferred to his friend's account = 41000

$\Rightarrow (20 + 21) \text{ unit} = 41000$

$\Rightarrow 41 \text{ unit} = 41000$

$\Rightarrow 1 \text{ unit} = 1000$

$\Rightarrow 50 \text{ unit} = 50000$

$\therefore$ The monthly salary of Shiva is Rs. 50000.

$\therefore$ The correct answer is option (c).

8. (a) Given,

Students who were successfully appeared = 78%

Total number of failure = 176

Students who got first class = 34%

Students who failed = $(100-78)\% = 22\%$

22% of total students = 176

1% of total students = $\frac{176}{22} = 8$

34% of total students = $8 \times 34 = 272$

The number of students who got first class = 272

$\therefore$ The correct answer is option (a).

9. (b) Let the initial value of fraction be x .

Then new value of fraction = $\frac{5}{4}$

$\therefore \frac{5}{4} = x \times \frac{125}{0.8}$

$\therefore \frac{5}{4} = x \times \frac{25}{16}$

$z = \frac{5}{4} \times \frac{16}{25} \Rightarrow x = \frac{4}{5}$

$\therefore$ The correct answer is option (b).

10. (d) Let the total student be x .

Given that, 60% students passed in half yearly examination.

Then, passed student in half yearly

$Y = \frac{60}{100} \times x = 0.60x$

Out of these passed in half yearly, 70% students passed in annual exam.

Then passed students in annual examination



$$= \frac{70}{100} \times 0.60x = 0.42x$$

The number of students who did not pass in half yearly examination

$$= x - 0.60x = 0.40x$$

Given, out of these who didn't pass the half yearly exam, 80% students passed in annual exam.

Then out of these students who did not pass in half-yearly examination, number of students who passed in annual exam =

$$\frac{80}{100} \times 0.40x + 0.32x = 0.74x$$

Therefore, percentage of students who passed annual

$$\text{Exam} = \frac{0.74x}{x} \times 100 = 74\%$$

Hence, 74% students passed the annual exam.

11. (c) Let the salary of Hemant be Rs. X.

He deposit 10% in PF.

$$\text{Remaining} = x - \frac{10x}{100} = \frac{90x}{100}$$

He saves 30% of remaining

$$\text{Saving of Hemant} = \frac{30}{100} \times \frac{90x}{100} = \frac{27x}{100}$$

$$\begin{aligned} \text{Expenses of Hemant} &= \frac{90x}{100} - \frac{27x}{100} \\ &= \frac{63x}{100} \dots\dots\dots(i) \end{aligned}$$

The ratio of expenses of medicine and groceries is 3:4 and the expenses of medicine is Rs. 2700.

$$\Rightarrow 3 \text{ units} = 2700$$

$$\Rightarrow 1 \text{ unit} = 900$$

$$\Rightarrow 4 \text{ units} = 3600$$

$$\begin{aligned} \text{Total expenses} &= 2700 + 3600 \\ &= 6300 \end{aligned}$$

From Eqs. (i) and (ii)(ii)

$$\Rightarrow \frac{63x}{100} = 6300$$

$$\Rightarrow x = 10000$$

So Monthly salary of Hemant is Rs. 10000.

